25) $f(x) = x^4 + x^2 + 1, \quad a = -2$

$f(a) = (-2)^4 + (-2)^2 + 1 = 21$

$f'(a) = 4x^3 + 2x = -36$

$f''(a) = 12x^2 + 2 = 50$

$f'''(a) = 24x = -48$

$f''''(a) = 24$

$$\sum_{n=0}^{\infty} \frac{f^{n}(a)}{n!}(x-a)^n = 21 - 36(x+2) + 25(x+2)^2 - 8(x+2)^3 + (x+2)^4.$$